

CHE 201 Fall 2019 HW 4

Kayla Klein

TOTAL POINTS

62.5 / 100

QUESTION 1

11 (3.8) 3.5 / 15

- **0 pts** Correct
- ✓ - **5 pts** Missing or wrong sketch of the phase diagram with allocation of points a, b and c
 - **0.5 pts** Wrong (T) for the triple point
 - **1.5 pts** Wrong location for a) in the phase diagram
 - **1.5 pts** Wrong location for b) in the phase diagram
 - **1.5 pts** Wrong location for c) in the phase diagram
- **5 pts** Wrong sketch of P-v diagram
- **1 pts** Wrong location for a) on the T-v diagram
- ✓ - **1 pts** Wrong or missing location for b) on the T-v diagram
- ✓ - **1 pts** Wrong or missing location for c) on the T-v diagram
 - **1 pts** Wrong or missing the location of point a) on the P-v diagram
 - **1 pts** Wrong or missing location for b) on the P-v diagram
 - **1 pts** Wrong or missing location for c) on the P-v diagram
- ✓ - **4.5 pts** Missing P-v diagram for point a) and T-v diagram for b) and c)
 - **0.5 pts** Wrong temperature line on P-v diagram for a)
 - **15 pts** Wrong answer
 - **5 pts** Wrong sketch of T-v diagram
 - **1 pts** Missing temperature line on P-v diagram
 - **3.5 pts** Late submission (50% off)
 - **6.5 pts** Late submission (50% off)

QUESTION 2

2 2 (3.12) 15 / 15

- ✓ - **0 pts** Correct
- **2.5 pts** Correct equation, but wrong solution for

the volume fraction. Correct answer is 97.7%

- **15 pts** Illegible answer
- **2.5 pts** $v_g = 0.0236$ (it's not $\times 10^{-3}$)
- **5 pts** Did not calculate the volume percentage
- **5 pts** Wrong calculation for V
- **5 pts** Wrong or missing calculation for V_g
- **7.5 pts** Late submission (50% off)
- **15 pts** Wrong solution

QUESTION 3

3 3 (3.25) 14 / 20

- **0 pts** Correct
- ✓ - **3 pts** $v_{g1} = 0.0121 \text{ m}^3/\text{kg}$ @ 16 bar (Table A-11)
- ✓ - **3 pts** $v_2 = 0.02017 \text{ m}^3/\text{kg}$ @ 160 oC (Table A-12)
 - **20 pts** Wrong solution
 - **2.5 pts** Correct equation, but wrong solution for work. $W = 12.9 \text{ kJ/kg}$
 - **7 pts** Late submission (50% off)
 - **10 pts** Late submission (50% off)
 - **10 pts** Missing or wrong work calculation
 - **2.5 pts** The assumption that $Q = \Delta H$ is valid for this case, but it doesn't come directly from the energy balance equation and it's not a general equation

QUESTION 4

4 4 (3.35) 10 / 15

- **0 pts** Correct
- ✓ - **5 pts** Wrong or missing sketch of T-v diagram
 - **2 pts** What is 872.6 Btu/lb?
 - **2.5 pts** Wrong temperature for the critical point of water. $T_c = 705.4 \text{ oF}$
 - **3 pts** Wrong sketch for the process (the transformation occurs at constant total volume and point 2 is inside the L+V region).
- **10 pts** Wrong heat transfer calculation
- **5 pts** Wrong value for x or missing calculation.

Correct value is $x = 0.0173$

- **3 pts** Missing stage 2 on sketch of T-v diagram
- **5 pts** Wrong solution for u_2
- **15 pts** Missing answer
- **7.5 pts** Late submission (50% off)

QUESTION 5

5 5 (3.44) 20 / 20

✓ - **0 pts** Correct

- **20 pts** This is not the ans for Q5
- **5 pts** $Q = \Delta U + W$ which W is negative
- **10 pts** U_1 and U_2 incorrect
- **5 pts** W incorrect
- **5 pts** U_2 incorrect
- **8 pts** $Q = ?$

QUESTION 6

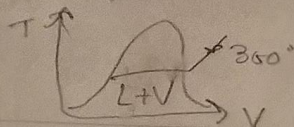
6 6 (3.9D) 0 / 15

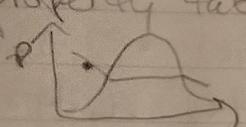
- **3 pts** Reference
- **15 pts** wrong submission
- **0 pts** Click here to replace this description.
- **15 pts** No submission
- ✓ - **15 pts** unable to read and no reference

Kaufa
Klein

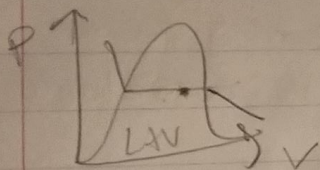
Homework #4

(3.8, 3.12, 3.25, 3.35, 3.44, 3.90)

- 1) a) property table $\rightarrow P = 20 \frac{\text{lb}}{\text{in}^2}$ $T_{\text{sat}} = 227.96^\circ\text{F}$ $T > T_{\text{sat}} \rightarrow$ superheated vapor phase
- 

- b) property table $\rightarrow$ at $P = 90 \frac{\text{lb}}{\text{in}^2}$ $T_{\text{sat}} = 320.31^\circ\text{F}$ $T < T_{\text{sat}}$
water will be liquid
- 

- c) $v_g = 6.472 \frac{\text{ft}^3}{\text{lb}}$ $v_f = .01745 \frac{\text{ft}^3}{\text{lb}}$ $v_f < v < v_g \rightarrow$ water in liq-vap mix
 $5 = .01745 + x(6.472 - .01745) \rightarrow x = .772$



- 2) $P = 10 \text{ bar}$ $v_f = .8352 \times 10^{-3} \text{ m}^3/\text{kg}$ $v_g = .0236 \text{ m}^3/\text{kg}$
 $v = v_f + x(v_g - v_f)$ $v = .8352 \times 10^{-3} + .6 \times (.0236 - .8352 \times 10^{-3})$
 $v = .01449 \text{ m}^3/\text{kg}$ $.6 = \frac{m_g}{m_g + 25} \rightarrow m_g = 37.5 \text{ kg}$ $v_g = .0236 \times 37.5 = .885 \text{ m}^3$
 $V_L = 25 \times .8352 \times 10^{-3} = .02088 \text{ m}^3$ $V = m v$ $V = (25 + 37.5) \times .01449 = .9056 \text{ m}^3$
 $x_{\text{vapor}} = \frac{v_g}{v} = \frac{.885}{.9056} \times 100$ $x_{\text{vapor}} = 97.725\%$

- 3) $P = \frac{F}{A} = \frac{471.1}{\pi \times .022} = 1499557.8738 \text{ Pa}$ $P_{\text{atm}} = 1 \text{ bar} = 1 \times 10^5 \text{ Pa}$
 $P = P_{\text{atm}} + P_{\text{piston}} = (1 \times 10^5) + (1499557.8738) = 15.99557 \times 10^5 \text{ Pa}$
 $16 \times 10^5 \text{ Pa} = 16 \text{ bar} \rightarrow V_1 = V_2 = .0123 \text{ m}^3/\text{kg}$
At 160°C + 16 bar $v_2 = .02019 \text{ m}^3/\text{kg}$ $w = P(v_2 - v_1)$
 $w = (1600 \text{ kPa} \times .02019 - .0123) = 12.624 \text{ kJ/kg}$

- 4) $\Delta E = \Delta U + \Delta KE + \Delta PE = 0 - 0 + 0 \rightarrow \Delta U = 0$ $Q/m = \Delta u$
 $T_c = 705.4^\circ\text{F}$ $P_c = 3204 \frac{\text{lb}}{\text{in}^2}$ $v_c = .05053 \frac{\text{ft}^3}{\text{lb}}$ $v_c = .0123 \frac{\text{ft}^3}{\text{lb}}$
closed system so specific vol. stays same $v_1 = v_2$
at $T = 400^\circ\text{F}$ the water becomes 2 phase same $v_1 = v_2$
 $v_g = 1.866 \frac{\text{ft}^3}{\text{lb}}$ $v_f = 374.3 \frac{\text{ft}^3}{\text{lb}}$ $u_g = 116.6 \frac{\text{Btu}}{\text{lb}}$ $u_f = .0123 \frac{\text{ft}^3}{\text{lb}}$
 $x = \frac{v - v_f}{v_g - v_f} = \frac{.05053 - .01864}{1.866 - .01864} = .0173$ $u = u_f + x(u_g - u_f)$
 $u = 374.3 + (.0173 \times 116.6 - 374.3) = 387.14 \frac{\text{Btu}}{\text{lb}}$ $Q/m = \Delta u = (387.14 - 872.16) = -485.5 \frac{\text{Btu}}{\text{lb}}$
 $Q/m = (387.14 - 872.16) = -485.5 \frac{\text{Btu}}{\text{lb}}$ $u_g = 223.02 \frac{\text{Btu}}{\text{lb}}$ $v_g = .1606 \times 10^{-3} \frac{\text{m}^3}{\text{kg}}$
 $Q/m = 33.27 \frac{\text{kJ}}{\text{kg}}$ $u_g = 223.02 \frac{\text{kJ}}{\text{kg}}$ $v_g = .1606 \times 10^{-3} \frac{\text{m}^3}{\text{kg}}$

11 (3.8) 3.5 / 15

- 0 pts Correct

✓ - 5 pts Missing or wrong sketch of the phase diagram with allocation of points a, b and c

- 0.5 pts Wrong (T) for the triple point

- 1.5 pts Wrong location for a) in the phase diagram

- 1.5 pts Wrong location for b) in the phase diagram

- 1.5 pts Wrong location for c) in the phase diagram

- 5 pts Wrong sketch of P-v diagram

- 1 pts Wrong location for a) on the T-v diagram

✓ - 1 pts Wrong or missing location for b) on the T-v diagram

✓ - 1 pts Wrong or missing location for c) on the T-v diagram

- 1 pts Wrong or missing the location of point a) on the P-v diagram

- 1 pts Wrong or missing location for b) on the P-v diagram

- 1 pts Wrong or missing location for c) on the P-v diagram

✓ - 4.5 pts Missing P-v diagram for point a) and T-v diagram for b) and c)

- 0.5 pts Wrong temperature line on P-v diagram for a)

- 15 pts Wrong answer

- 5 pts Wrong sketch of T-v diagram

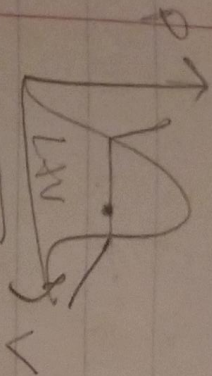
- 1 pts Missing temperature line on P-v diagram

- 3.5 pts Late submission (50% off)

- 6.5 pts Late submission (50% off)

B) Property table $\rightarrow$ at $P = 90 \text{ lbf/in}^2$ $T_{\text{sat}} = 320.31^\circ\text{F}$ $T < T_{\text{sat}}$
 water will be dry saturated

C) $V_g = 6.472 \text{ ft}^3/\text{lb}$ $V_f = 0.01745 \text{ ft}^3/\text{lb}$ $V_f < V < V_g \rightarrow$ water in dry-veg mix
 $S = 0.01745 + x \cdot 6.472 - 0.01745 \rightarrow x = 0.772$



2) $P = 10 \text{ bar}$ $V_f = 1.8352 \times 10^{-3} \text{ m}^3/\text{kg}$ $V_g = 0.0236 \text{ m}^3/\text{kg}$
 $V = 0.01449 \text{ m}^3/\text{kg}$ $V = V_f + x(V_g - V_f)$ $x = 0.8352 \times 10^{-3} + 0.6 \times (0.0236 - 0.8352 \times 10^{-3})$
 $V_g = 25 \times 8352 \times 10^{-3} = 0.2088 \text{ m}^3$ $V = m_g V_g$ $V_g = 0.0236 \times 37.5 = 0.885 \text{ m}^3$
 $V = (25 + 37.5) \times 0.01449$

$x_{\text{vapor}} = \frac{V_g}{V} = \frac{0.885}{0.056} \times 100$ $x_{\text{vapor}} = 97.725\%$
 $P = \frac{F}{A} = \frac{411.1}{\pi \times 0.022} = 1499557.8738 \text{ Pa}$ $P_{\text{atm}} = 1 \text{ bar} = 1 \times 10^5 \text{ Pa}$
 $P = P_{\text{atm}} + P_{\text{piston}} = (1 \times 10^5) + (1499557.8738) = 15.99557 \times 10^5 \text{ Pa}$

3) $P = 10 \text{ bar}$ $V_1 = V_g = 0.0123 \text{ m}^3/\text{kg}$
 $u_{10} \times 10^5 \text{ Pa} = 16 \text{ bar} \rightarrow V_1 = V_g = 0.0123 \text{ m}^3/\text{kg}$ $u = P(V_2 - V_1)$

At 1600°C $V_2 = 0.2019 \text{ m}^3/\text{kg}$ $u = P(V_2 - V_1)$

$u = (1600 \text{ kPa}) \cdot 0.2019 - 0.0123 = 12.624 \text{ kPa/kg}$

4) $\Delta E = \Delta U + \Delta KE + \Delta PE = Q - W \rightarrow Q - W = 0 \rightarrow \Delta U = Q$ $Q/m = \Delta u$
 $T_c = 705.4^\circ\text{C}$ $P_c = 3204 \text{ lbf/in}^2$ $V_c = 0.05053 \text{ ft}^3/\text{lb}$ $V_c = 872.6 \text{ cm}^3/\text{lb}$

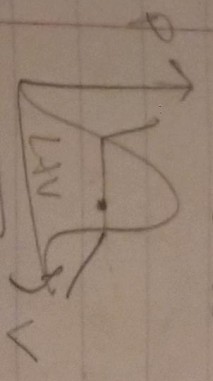
- closed system so specific vol. stays same $u_1 = u_2$
 at $T = 400^\circ\text{C}$ the water becomes 2 phase $u_f = 0.01864 \text{ ft}^3/\text{lb}$

2 2 (3.12) 15 / 15

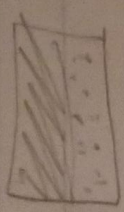
✓ - 0 pts Correct

- 2.5 pts Correct equation, but wrong solution for the volume fraction. Correct answer is 97.7%
- 15 pts Illegible answer
- 2.5 pts $v_g = 0.0236$ (it's not $\times 10^{-3}$)
- 5 pts Did not calculate the volume percentage
- 5 pts Wrong calculation for V
- 5 pts Wrong or missing calculation for V_g
- 7.5 pts Late submission (50% off)
- 15 pts Wrong solution

$$S = .01745 + x(16.472 - .81745) \rightarrow x = .772$$



2)



$P = 10 \text{ bar}$ $V_f = 1.8352 \times 10^{-3} \text{ m}^3/\text{kg}$ $V_g = .0236 \text{ m}^3/\text{kg}$

$V = .01449 \text{ m}^3/\text{kg}$ $V = V_f + x(V_g - V_f)$ $V = .8352 \times 10^{-3} + .6 \times (.0236 - .8352 \times 10^{-3})$

$V_f = 25 \times .8352 \times 10^{-3} = .02088 \text{ m}^3$ $.6 = \frac{m_g}{m_g + 25} \rightarrow m_g = 37.5 \text{ kg}$ $V_g = .0236 \times 37.5 = .885 \text{ m}^3$

$X_{\text{vapor}} = \frac{V_g}{V} = \frac{.885}{.9056} \times 100$ $X_{\text{vapor}} = 97.725\%$ $X_{\text{vapor}} = 97.725\%$

3) $P = \frac{F}{A} = \frac{411.1}{\pi \times .022} = 1499557.8738 \text{ Pa}$ $P_{\text{atm}} = 1 \text{ bar} = 1 \times 10^5 \text{ Pa}$

$P = P_{\text{atm}} + P_{\text{piston}} = (1 \times 10^5) + (1499557.8738) = 15.99557 \times 10^5 \text{ Pa}$

$1016 \times 10^5 \text{ Pa} = 16 \text{ bar} \rightarrow V_1 = V_g = .0123 \text{ m}^3/\text{kg}$

At 1600°C $V_2 = .02019 \text{ m}^3/\text{kg}$ $W = P(V_2 - V_1)$

$W = (1600 \text{ kPa}) \cdot .02019 - (.0123) = 12.624 \text{ kPa/kg}$

4)

$\Delta E = \Delta U + \Delta KE + \Delta PE = Q - W \rightarrow Q - W = 0 \rightarrow \Delta U = Q$ $Q/m = \Delta U$

$T_c = 705.4^\circ\text{F}$ $P_c = 3264 \text{ lbf/in}^2$ $V_c = .05053 \text{ ft}^3/\text{lb}$ $V_c = 872.6 \text{ cm}^3/\text{lb}$

- closed system so specific vol. stays same $v_2 = v_c$

at $T = 400^\circ\text{F}$ the water became 2 phase $v_f = .01864 \text{ ft}^3/\text{lb}$

$v_g = 1.866 \text{ ft}^3/\text{lb}$ $v_f = 374.3 \text{ cm}^3/\text{lb}$ $v_g = 116.6 \text{ cm}^3/\text{lb}$

$x = \frac{v - v_f}{v_g - v_f} = \frac{.05053 - .01864}{1.866 - .01864} = .0173$ $U = 374.3 + (.0173)(116.6 - 374.3) = 387.14 \text{ Btu/lb}$ $Q/m = \Delta U = (U_2 - U_1)$

$Q/m = (387.14 - 872.6) = -485.5 \text{ Btu/lb}$

5)

$T_1 = -10^\circ\text{C}$ $U_f = 33.27 \text{ kJ/kg}$ $U_g = 223.02 \text{ kJ/kg}$ $U_1 = U_f + x(U_g - U_f)$

3 3 (3.25) 14 / 20

- 0 pts Correct
- ✓ - 3 pts $v_{g1} = 0.0121 \text{ m}^3/\text{kg}$ @ 16 bar (Table A-11)
- ✓ - 3 pts $v_2 = 0.02017 \text{ m}^3/\text{kg}$ @ 160 oC (Table A-12)
- 20 pts Wrong solution
- 2.5 pts Correct equation, but wrong solution for work. $W = 12.9 \text{ kJ/kg}$
- 7 pts Late submission (50% off)
- 10 pts Late submission (50% off)
- 10 pts Missing or wrong work calculation
- 2.5 pts The assumption that $Q = \Delta H$ is valid for this case, but it doesn't come directly from the energy balance equation and it's not a general equation

3) $P = \frac{F}{A} = \frac{411.1}{.0056} = 73,380 \text{ Pa}$ $(X_{\text{water}} = 97.725\%)$

$P_{\text{atm}} = 1 \times 10^5 \text{ Pa}$

$P_{\text{position}} = (1 \times 10^5) + (1499557.8738) = 15.99557 \times 10^5 \text{ Pa}$

At 1600°C + 16 bar $V_2 = 0.2019 \text{ m}^3/\text{kg}$ $u = P(V_2 - V_1)$

$u = (1600 \text{ kPa} \times 0.2019 - 0.0123) = 12.624 \text{ kJ/kg}$

4) $\Delta E = \Delta U + \Delta KE + \Delta PE = Q - W > 0 \rightarrow \Delta U = Q$ $Q/m = \Delta u$

$T_c = 705.4^\circ\text{C}$ $P_c = 3264 \text{ kPa}/\text{m}^2$ $V_c = 0.5053 \text{ m}^3/\text{kg}$ $V_c = 872.6 \text{ cm}^3/\text{kg}$

- Closed System so specific vol. stays same $v_2 = v_c$
 at $T = 400^\circ\text{C}$ the water becomes 2 phase $v_f = 0.01864 \text{ m}^3/\text{kg}$

$v_g = 1.846 \text{ m}^3/\text{kg}$ $v_f = 374.3 \text{ cm}^3/\text{kg}$ $u_g = 1116.6 \text{ kJ/kg}$

$x = \frac{v - v_f}{v_g - v_f} = \frac{0.5053 - 0.01864}{1.846 - 0.01864} = 0.0173$

$u = 374.3 + (0.0173)(1116.6 - 374.3) = 387.14 \text{ kJ/kg}$ $Q/m = \Delta u = (u_2 - u_1)$

5) $T_1 = -10^\circ\text{C}$ $v_f = 33.27 \text{ cm}^3/\text{kg}$ $u_g = 223.02 \text{ kJ/kg}$ $u_1 = u_f + x(u_g - u_f)$

$u_1 = 33.27 + .8(223.02 - 33.27) = 185.07 \text{ kJ/kg}$ $v_f = 1606 \times 10^{-3} \text{ m}^3/\text{kg}$

$v_g = 0.052 \text{ m}^3/\text{kg}$ $v_1 = v_f + x(v_g - v_f) = .7606 \times 10^{-3} + .8(.052 - .7606 \times 10^{-3})$

$v_1 = 0.0523 \text{ m}^3/\text{kg}$ $v_2 = v_1 = .0523 \text{ m}^3/\text{kg}$

$T_2 = 40^\circ\text{C}$ and $v_2 = .052312 \text{ m}^3/\text{kg}$ $u_2 = 250.380 \text{ kJ/kg}$

$m = \frac{V}{v} = \frac{0.01 \text{ m}^3}{0.0523 \text{ m}^3/\text{kg}} = .1912 \text{ kg}$ $W = -V_1 \Delta P = -(12 \text{ V} \times 5 \text{ A}) \times 5 \text{ min} \times \frac{1 \text{ W}}{1 \text{ V} \cdot \text{A}} \times \frac{60 \text{ sec}}{1 \text{ min}} \times \frac{1 \text{ kJ}}{1000 \text{ J}} = -18 \text{ kJ}$

4 4 (3.35) 10 / 15

- 0 pts Correct

✓ - 5 pts Wrong or missing sketch of T-v diagram

- 2 pts What is 872.6 Btu/lb?

- 2.5 pts Wrong temperature for the critical point of water. $T_c = 705.4$ oF

- 3 pts Wrong sketch for the process (the transformation occurs at constant total volume and point 2 is inside the L+V region).

- 10 pts Wrong heat transfer calculation

- 5 pts Wrong value for x or missing calculation. Correct value is $x = 0.0173$

- 3 pts Missing stage 2 on sketch of T-v diagram

- 5 pts Wrong solution for u_2

- 15 pts Missing answer

- 7.5 pts Late submission (50% off)

$$\omega = (1600 \text{ kPa} \times 0.02019 - 0.0123) = 12.624 \text{ kPa/kg}$$

$$4) \Delta E = \Delta U + \Delta KE + \Delta PE = 0 - \dot{W} \rightarrow 0 \rightarrow \Delta U = 0 \quad Q/m = \Delta u$$

$$T_c = 705.4^\circ\text{F} \quad p_c = 3204 \text{ lbf/in}^2 \quad v_c = 0.05053 \text{ ft}^3/\text{lb} \quad u_c = 872.6 \text{ Btu/lb}$$

close to system so specific vol. stays same $v_2 = v_c$

at $T = 400^\circ\text{F}$ the water becomes 2 phase $v_f = 0.01864 \text{ ft}^3/\text{lb}$

$$u_g = 1.866 \text{ Btu/lb} \quad u_f = 374.3 \text{ Btu/lb} \quad u_g = 1116.6 \text{ Btu/lb}$$

$$x = \frac{v - v_f}{v_g - v_f} = \frac{0.05053 - 0.01864}{1.866 - 0.01864} = 0.0173 \quad u = u_f + x(u_g - u_f)$$

$$u = 374.3 + (0.0173 \times 1116.6 - 374.3) = 387.14 \text{ Btu/lb} \quad Q/m = \Delta u = (u_2 - u_1)$$

$$Q/m = (387.14 - 872.6) = -485.5 \text{ Btu/lb}$$

$$5) T_1 = -10^\circ\text{C} \quad u_f = 33.27 \text{ kJ/kg} \quad u_g = 223.02 \text{ kJ/kg} \quad u_1 = u_f + x(u_g - u_f)$$

$$u_1 = 33.27 + .8(223.02 - 33.27) = 185.07 \text{ kJ/kg} \quad v_f = .7606 \times 10^{-3} \text{ m}^3/\text{kg}$$

$$v_g = .0652 \text{ m}^3/\text{kg} \quad v_1 = v_f + x(v_g - v_f) = .7606 \times 10^{-3} + .8(.0652 - .7606 \times 10^{-3})$$

$$v_1 = .0523 \text{ m}^3/\text{kg} \quad v_2 = v_1 = .0523 \text{ m}^3/\text{kg}$$

$$T_2 = 40^\circ\text{C} \quad \text{and} \quad v_2 = .052312 \text{ m}^3/\text{kg} \quad u_2 = 250.380 \text{ kJ/kg}$$

$$m = \frac{v}{v_1} = \frac{0.01 \text{ m}^3}{.0523 \text{ m}^3/\text{kg}} = .1912 \text{ kg} \quad W = -V_1 \Delta T = -(12 \text{ V} \times 5 \text{ A} \times 5 \text{ min}) \times \frac{1 \text{ W/A} \times 60 \text{ sec}}{1 \text{ V}} \times \frac{1 \text{ min}}{1 \text{ W}} \times \frac{1 \text{ kJ}}{10^3 \text{ J}} = -18 \text{ kJ}$$

5 5 (3.44) 20 / 20

✓ - 0 pts Correct

- 20 pts This is not the ans for Q5
- 5 pts $Q = \Delta U + W$ which W is negative
- 10 pts U_1 and U_2 incorrect
- 5 pts W incorrect
- 5 pts U_2 incorrect
- 8 pts $Q = ?$

$$Q = -5.51 \times 5$$

$$Q = -m(0.2 - 0.1) + W = .1912 \text{ kg} (250 \cdot 38 - 185 \cdot 0.7) \text{ kg} / \text{kg} - 13 \text{ kg}$$

6) My use 1

Shower - 30 mins $\times 2$ wash dishes - 20 mins
 drink water - 8 glasses $\times 3$ brush teeth 2 mins $\times 6$
 make needles $\times 2$

1 gallon wouldn't be enough to wash my dishes or shower.
 if also wouldn't be enough to drink and stay hydrated properly

- can improve water usage by cutting off shower 3 zone
 and getting what chemicals we use to wash ourselves
 and our dishes

- we can mandate what companies dump into the water

Water usage in USA

www.epa.gov $\rightarrow$ 20% Shower, 24% toilet, 8% other, 12% leak,
 17% clothes wash, 19% faucet

enrpa.com $\rightarrow$ America uses 322 billion gallons of water each day

usgs.gov $\rightarrow$ USA uses water for irrigation, public supply,
 domestic, industrial, mining, livestock, thermoelectric power

6 6 (3.9D) 0 / 15

- 3 pts Reference
- 15 pts wrong submission
- 0 pts [Click here to replace this description.](#)
- 15 pts No submission
- ✓ - 15 pts unable to read and no reference